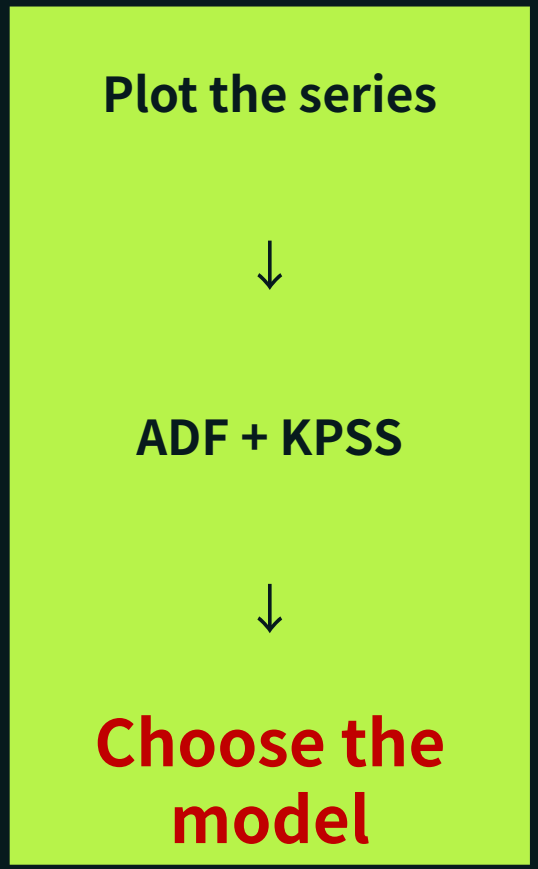
Stationarity and Unit Roots

The starting point of time-series analysis



Four steps lead from definitions to tests and finance cases

01	**Foundations**	**What is stationarity?** Weak stationarity · I(d) · why it matters
02	**AR(1) and unit roots**	**Do shocks persist or fade?** Mean reversion · half-life · TS vs DS
03	**Unit root tests**	**Guarding against spurious regression** DF · ADF · KPSS · analysis workflow
04	**Finance cases and lab**	**Prices · FX · volatility · rates** Spurious regression · PPP · Claude Code

Plot → test → choose levels, differences or VECM

A stationary series has a stable mean, variance and autocovariance

Weak (covariance) stationarity — what practitioners mean by 'stationary'

① Constant mean

$$\mathrm{E}(y_t) = m$$

② Constant variance

$$\mathrm{Var}(y_t) = \sigma^2$$

③ Covariance by lag only

$$\mathrm{Cov}(y_t,\, y_{t-k}) = C(k)$$

Returns to one level

Level does not wander

Same size of swings

Same width throughout

Dates do not matter

Only the lag matters

Forecasting works only if means and correlations estimated from the past still hold

Integration order counts the differences needed for stationarity

Difference = this period's value − last period's value

First difference

$$\Delta y_t = y_t - y_{t-1}$$

Strict stationarity

- Joint law is fixed
- Too strong to use

Integration order I(d)

- d diffs → I(0)
- RW is I(1)

Unit root

- A root equals 1
- AR(1): b = 1

A series with a unit root must be differenced once to become stationary

Ignoring a unit root undermines estimation, regression and testing

Three reasons to check stationarity before any analysis

01 · Meaning of estimates

Basis of forecasts

- Past means and
- correlations hold

02 · Spurious (1974)

'Significant' noise

- Independent RWs
- High R-squared
- Significant t

03 · Test distributions

t and F break down

- Usual tables fail
- Granger tests
- assume I(0)

Rule of thumb: suspect a spurious regression when R-squared exceeds Durbin–Watson

Check stationarity before anything else

When the AR(1) coefficient b equals 1, shocks never die out

b is the share of last period's value carried over; e is a zero-mean white-noise shock

AR(1) model

$$y_t = b\, y_{t-1} + e_t$$

|b| < 1 · mean reversion

$$|b| < 1: \quad \mathrm{Var}(y_t) = \frac{\sigma^2}{1-b^2}$$

b = 1 · variance grows

$$b = 1: \quad y_t = y_0 + e_1 + \cdots + e_t, \quad \mathrm{Var}(y_t) = t\,\sigma^2$$

|b| > 1

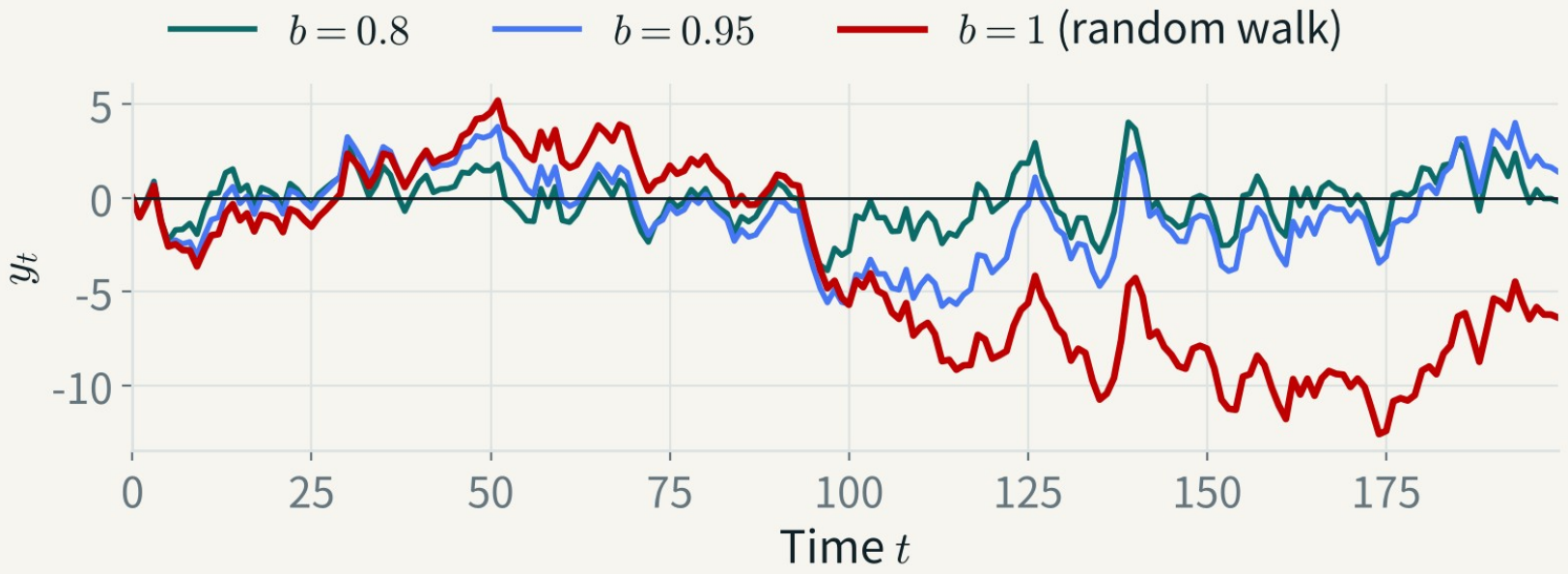
Explosive path

Covered by bubble tests

With b = 1, past shocks accumulate and the variance grows in proportion to time

As b approaches 1, even a stationary series wanders like a random walk

Simulation feeding the same shocks (seed 42, 200 periods) into b = 0.8 · 0.95 · 1



※ Simulated data — standard normal shocks, initial value 0

b = 0.8 returns to 0 quickly, 0.95 lingers for long, and 1 never returns

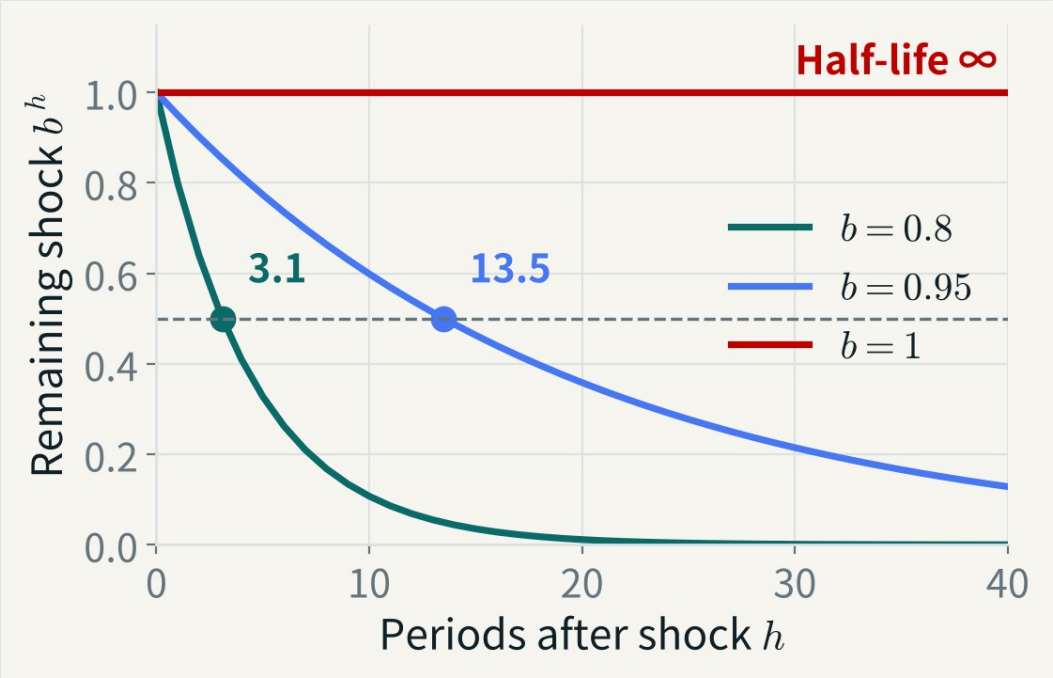
Shock half-lives spread from 3.1 to 13.5 periods to infinity

The point where the remaining share of a shock, b to the power h, falls to one half

Half-life (periods)

- b = 0.8 → 3.1
- b = 0.95 → 13.5
- b = 1 → ∞

Near 1, hard to tell from a unit root



This is why unit root tests have low power

Nonstationary series are trend stationary or difference stationary

a is the intercept (drift), c the time-trend slope, u a stationary fluctuation

Trend stationary (TS)

$$y_t = a + c\,t + u_t$$

Difference stationary (DS)

$$y_t = a + y_{t-1} + e_t \;\Rightarrow\; \Delta y_t = a + e_t$$

Detrending

- Remove trend → stationary
- Differencing → MA unit root

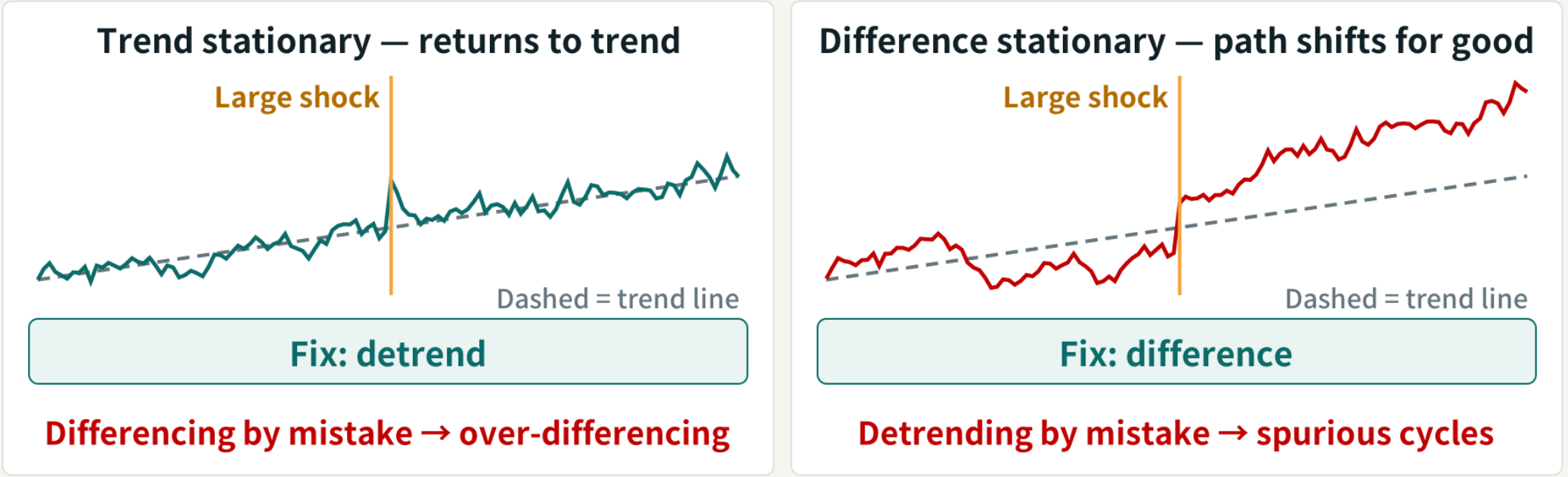
Differencing

- One difference → stationary
- Detrending → spurious cycles

Misjudging the type makes the treatment itself a new source of distortion

TS series return to trend, while DS paths shift permanently

Paths of the two models hit by a shock of the same size

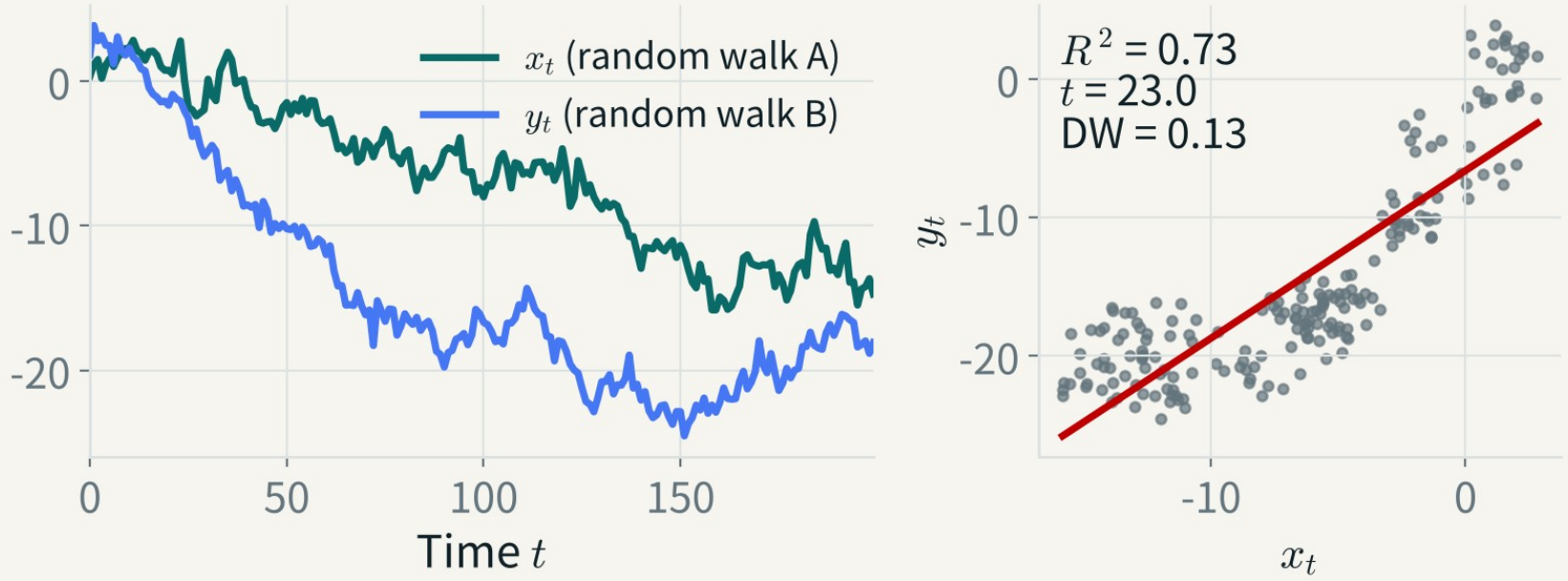


※ Simulated data (seed 11, 120 periods, large shock at t = 60)

Whether shocks are transitory or permanent separates the two types

Two unrelated random walks can show a strong regression fit

Levels regression of two independent random walks (seed 1, 200 periods)



※ Simulated data — share 'significant' at 5% over 1,000 runs: levels 81% · differences 5.5%

A high R-squared between levels may reflect unit roots rather than a real relationship

The Dickey–Fuller test checks if the lagged-level coefficient is zero

Regression obtained by subtracting the lagged value from both sides of the AR(1)

DF regression

$$\Delta y_t = (b-1)\, y_{t-1} + e_t, \qquad H_0 : b = 1 \quad H_1 : b < 1$$

5% CV · no constant	5% CV · constant	5% CV · constant + trend
−1.95	**−2.86**	**−3.41**

※ Large-sample values. Reject the unit root only if the t-stat is below (more negative than) these

The t-stat follows the DF distribution, so its critical values sit further left

The ADF test adds lagged differences to absorb serial correlation

Augmented Dickey–Fuller — the tested coefficient is the same lagged-level term as in DF

ADF regression

$$\Delta y_t = \underbrace{a + c\,t}_{\text{deterministic}} + \underbrace{(b-1)\,y_{t-1}}_{\text{test}} + \underbrace{d_1 \Delta y_{t-1} + \cdots + d_q \Delta y_{t-q}}_{\text{serial correlation}} + e_t$$

Lag length q

Chosen by AIC / BIC

Constant / trend

Decided from the plot

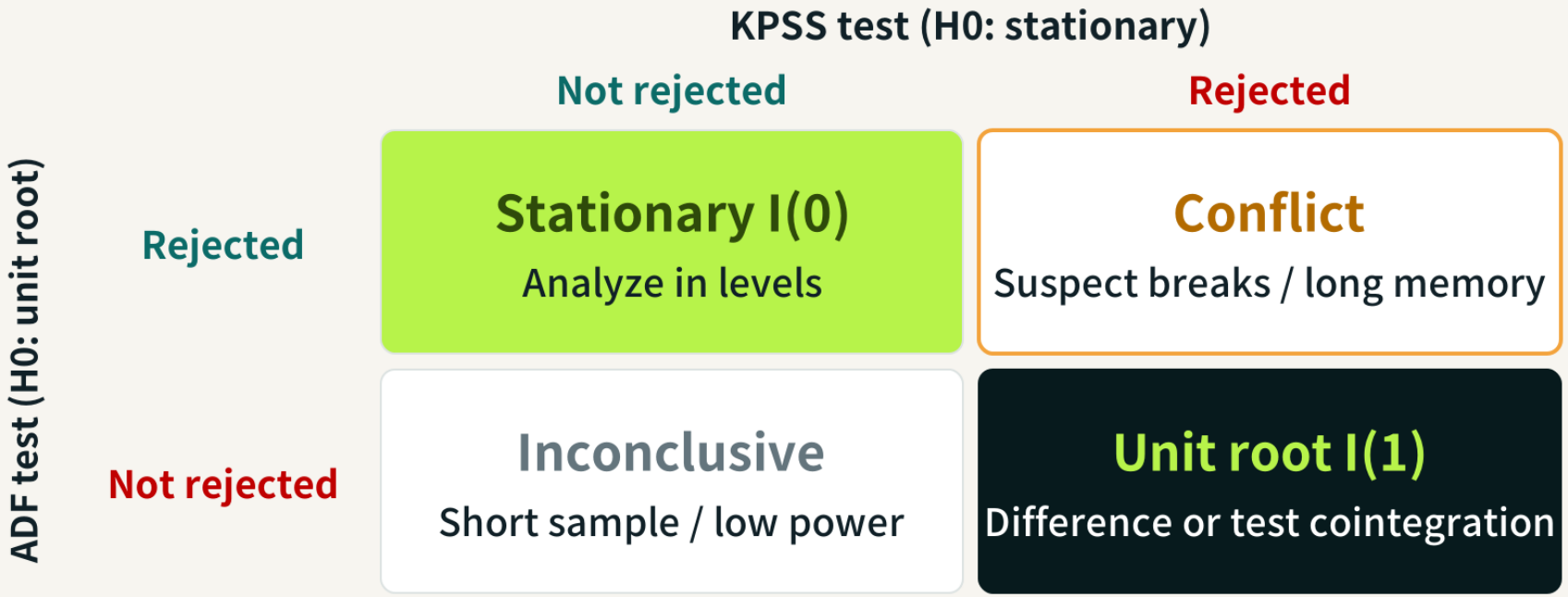
Alternative tests

PP · DF-GLS

The number of lags and the constant/trend choice can change the test result

Reading ADF and KPSS together makes the conclusion more robust

KPSS takes stationarity as the null, the reverse of ADF — read the two results in a 2×2 table



If they conflict, suspect structural breaks or long memory

If neither rejects, the verdict is inconclusive, not a unit root

Structural breaks and short samples mislead unit root tests

Three checks before taking a test result at face value

Structural break

I(0) looks like I(1)

- Perron (1989)
- Zivot–Andrews test
- Check crises

Sample length

Low power

- ~20 annual obs.
- rarely decide it

Interpretation

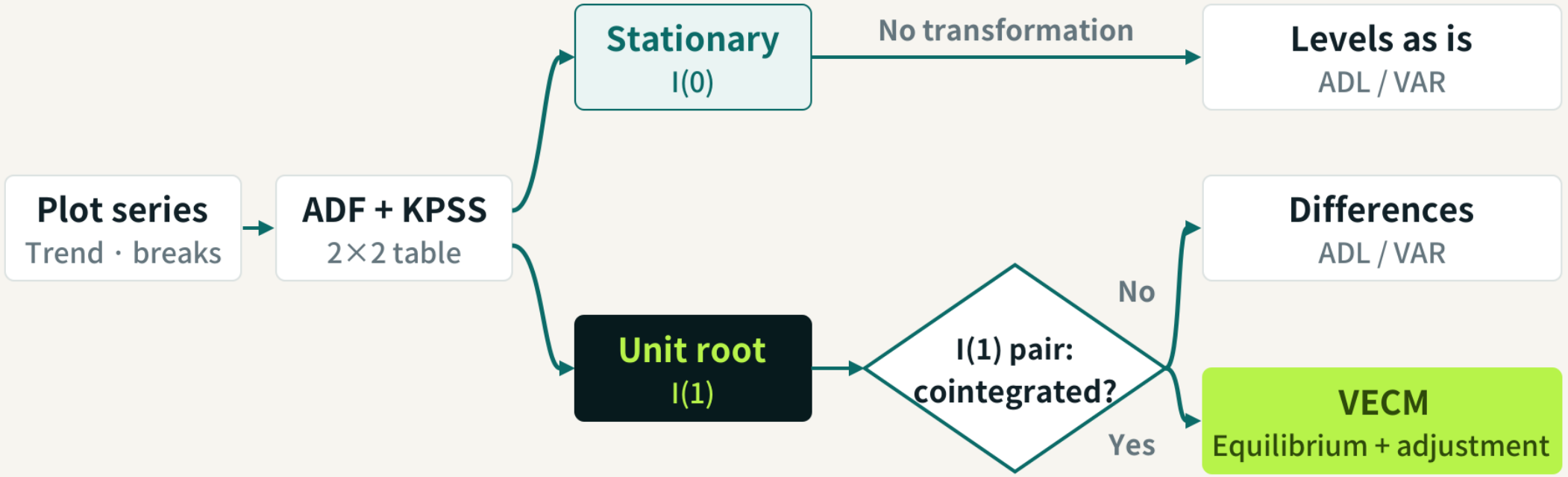
'Fail to reject'

- No proof of I(1)
- May be inconclusive

Rule of thumb: price, index and GDP levels are mostly I(1); returns and growth rates I(0)

Test results decide between levels, differences and a VECM

Stationary → keep levels; unit root → ask once more whether the series are cointegrated



If unit-root variables are cointegrated, a VECM keeps their long-run equilibrium

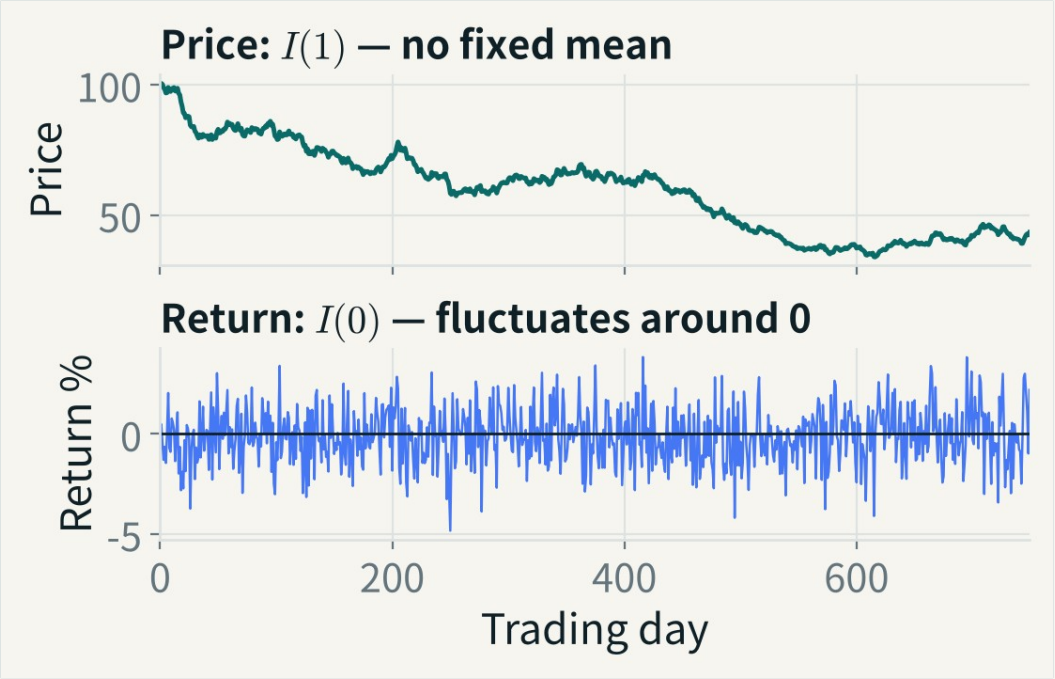
Price levels are treated as I(1) and returns as stationary

Case 1 · Stock price levels vs returns — the most basic split in financial time series

CASE 01 · Price regression trap

- Price-on-price OLS
- Spurious fit
- Use returns

Or test for cointegration



A high R-squared between two prices may be a spurious regression

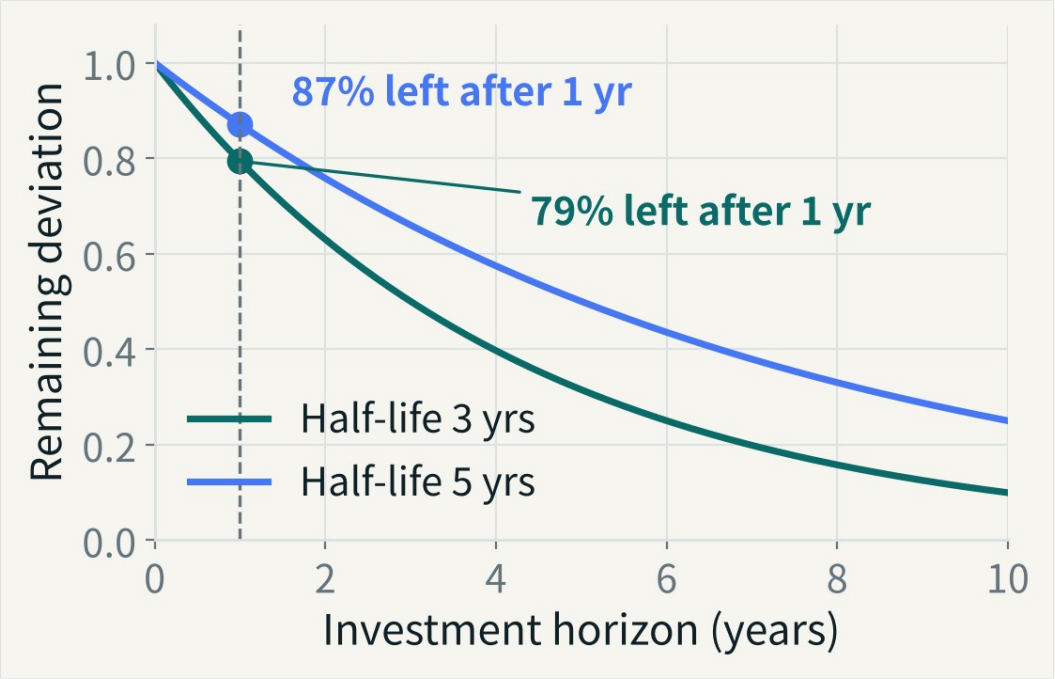
Mean reversion in real exchange rates is too slow for tests to detect

Case 2 · The PPP puzzle — half-life of 3 to 5 years (Rogoff, 1996)

CASE 02 · PPP puzzle

- Reverts in theory
- Tests can't reject
- Use panel tests

Classic case of low power



With a 3-year half-life, the rate is effectively a random walk to a 1-year investor

For highly persistent variables, tests and theory often disagree

Cases 3–5 · Variables close to a unit root but mean-reverting in theory

CASE 03 · D/P, CAPE

Stambaugh bias

- Inflated t-stats
- Overstated power
- Stambaugh (1999)

CASE 04 · Volatility (VIX)

Spikes, then reverts

- Highly persistent
- Half-life matters
- for vol and options

CASE 05 · Rate levels

Theory vs tests

- Bounded in theory
- Data: unit root
- State the rationale

Choose levels or differences from test results plus a stated purpose and rationale

The first lab tests the stationarity of prices and returns directly

Claude Code prompt ① — stationarity of prices and returns, plus a spurious regression experiment

Input data (yfinance)

Samsung Electronics · KOSPI daily close

Output

unit_root_lab.md

Download from 2010 → **Prices, returns ADF + KPSS** → **1,000 RW pairs spurious rate** → **Levels vs diffs regressions** → **3 lessons save outputs**

Students paste and check — Claude Code downloads, runs and fixes errors

The second lab compares VIX and real exchange rate half-lives

Claude Code prompt ② — comparing mean-reversion speed with FRED data

Input data (FRED)

VIXCLS · RBKRBIS monthly

Output

half_life.md

Download full history → ADF + KPSS 2×2 verdict → Fit AR(1) half-life (mo) → Zivot–Andrews pre/post break → Conclude by horizon

Use the half-life to judge which mean reversion is usable in a strategy

Stationarity underpins regression and testing.

A unit root makes shocks permanent.

Read ADF and KPSS together.

Next: cointegration and VECM —
long-run equilibrium of I(1) series